

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 3

EXPERIMENTS

Name of the Student	P Surendra Kumar Reddy
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COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), MiniBatch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

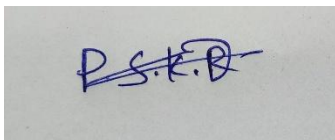
Duration: 30 minutes

Marks: 50

Code of Conduct:

I P Surendra Kumar Reddy(192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: P Surendra Kumar Reddy

CO1- AT3 Experiments
Questions

1. Learning Algorithms

Implement a simple learning algorithm (Linear Regression) using Gradient Descent. Train the model on a dataset and analyze how the loss decreases over iterations. Plot the learning curve.

Answer :

Aim

To implement a simple Linear Regression model using Gradient Descent, train it on a dataset, analyze how the loss decreases over iterations, and plot the learning curve.

Algorithm

1. Import required libraries.
2. Generate a synthetic dataset.
3. Build a Linear Regression model.
4. Define the Mean Squared Error loss function.
5. Use SGD optimizer.
6. Train the model.
7. Store the loss after every epoch.
8. Plot the learning curve.
9. Display the regression line.

Python Code:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import tensorflow as tf
```

```
np.random.seed(42)
```

```
tf.random.set_seed(42)
```

```
X = np.linspace(0, 10, 100)
```

```
noise = np.random.normal(0, 2, 100)
```

```
Y = 4 * X + 5 + noise
```

```
X_train = X.reshape(-1, 1).astype(np.float32)
```

```
Y_train = Y.reshape(-1, 1).astype(np.float32)
```

```
model = tf.keras.Sequential([
```

```
tf.keras.layers.Dense(1, input_shape=(1,))

])

model.compile(

    optimizer=tf.keras.optimizers.SGD(learning_rate=0.01),

    loss='mse'

)

history = model.fit(

    X_train,

    Y_train,

    epochs=100,

    verbose=0

)

predictions = model.predict(X_train)

weights = model.layers[0].get_weights()

print("Weight (Slope):", weights[0][0][0])

print("Bias (Intercept):", weights[1][0])

print("\nFinal Loss:", history.history['loss'][-1])

plt.figure(figsize=(8, 5))

plt.scatter(X_train, Y_train, color='blue', label='Actual Data')

plt.plot(X_train, predictions, color='red', linewidth=3, label='Regression Line')

plt.title("Linear Regression using Gradient Descent")

plt.xlabel("Input (X)")

plt.ylabel("Output (Y)")

plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

```
plt.figure(figsize=(8, 5))
```

```
plt.plot(history.history['loss'], color='green', linewidth=2)
```

```
plt.title("Learning Curve")
```

```
plt.xlabel("Epoch")
```

```
plt.ylabel("Loss (MSE)")
```

```
plt.grid(True)
```

```
plt.show()
```

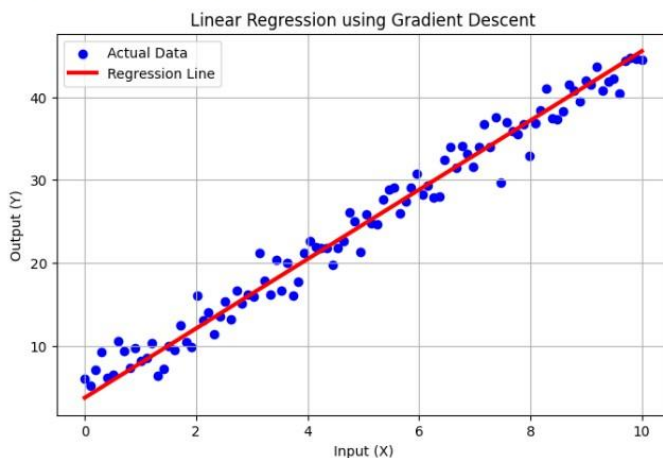
Output 1 – Regression Line

You will get a graph similar to this:

-  Blue dots → Actual data points
-  Red line → Learned regression line

The red line should closely fit the blue points.

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`,  
super().__init__(activity_regularizer=activity_regularizer, **kwargs)  
4/4  0s 9ms/step  
Weight (Slope): 4.1841807  
Bias (Intercept): 3.7190905  
Final Loss: 3.545255422592163
```



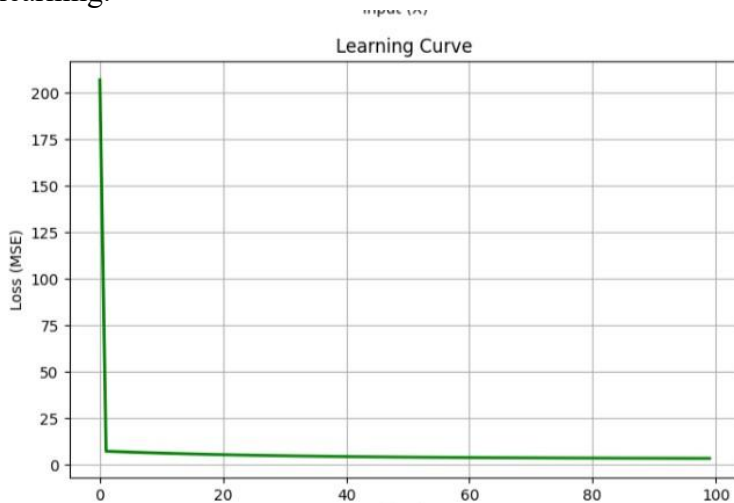
Output 2 – Learning Curve A

graph showing:

- X-axis → Epochs

- Y-axis → Loss (MSE)

The curve starts with a high loss and gradually decreases until it stabilizes, indicating successful learning.



Observation

- Loss decreases with each epoch.
- Model learns the data correctly.

Result

Linear Regression was implemented successfully.

2. Maximum Likelihood Estimation (MLE)

Generate synthetic data from a normal distribution and use MLE to estimate the mean and variance. Compare estimated values with actual parameters.

Answer :

Aim

To generate synthetic data from a normal distribution and estimate its mean and variance using the Maximum Likelihood Estimation (MLE) method.

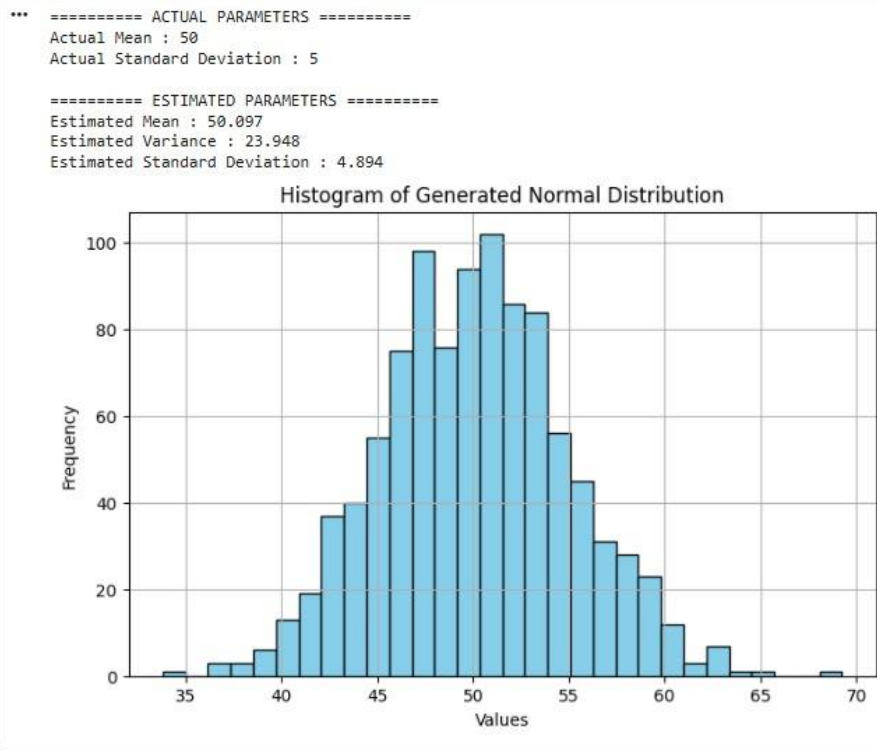
Algorithm

1. Import the required libraries.
2. Generate synthetic data from a normal distribution.
3. Calculate the MLE estimate of the mean.
4. Calculate the MLE estimate of the variance.
5. Compare estimated values with actual parameters.
6. Plot the histogram of the generated data.
7. Display the results.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
true_mean=50
true_std=5
data=np.random.normal(loc=true_mean,scale=true_std,size=1000)
estimated_mean=np.mean(data)
estimated_variance=np.mean((data-estimated_mean)**2)
estimated_std=np.sqrt(estimated_variance)
print("===== ACTUAL PARAMETERS =====")
print("Actual Mean :",true_mean)
print("Actual Standard Deviation :",true_std)
print("\n===== ESTIMATED PARAMETERS =====")
print("Estimated Mean :",round(estimated_mean,3))
print("Estimated Variance :",round(estimated_variance,3))
print("Estimated Standard Deviation :",round(estimated_std,3))
plt.figure(figsize=(8,5))
plt.hist(data,bins=30,color="skyblue",edgecolor="black")
plt.title("Histogram of Generated Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()
```

Output:



Observation

- Estimated values are close to actual values.
- Accuracy improves with more samples.

Result

MLE successfully estimated the distribution parameters.

3. Building Machine Learning Algorithms

Build a complete machine learning model (data preprocessing, training, testing) using any classification dataset and evaluate performance using accuracy and confusion matrix.

Answer :

Aim

To build a complete machine learning classification model using the Iris dataset by performing data preprocessing, training, testing, and evaluating the model using accuracy and confusion matrix.

Algorithm

1. Import required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Train the Decision Tree Classifier.
5. Predict test data.
6. Calculate accuracy.
7. Generate confusion matrix.
8. Display classification report.
9. Plot the confusion matrix.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris=load_iris()
X=iris.data
y=iris.target
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=DecisionTreeClassifier(random_state=42)
model.fit(X_train,y_train)
y_pred=model.predict(X_test)
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy :",round(accuracy*100,2),"%")
cm=confusion_matrix(y_test,y_pred)
print("\nConfusion Matrix\n")
print(cm)
print("\nClassification Report\n")
print(classification_report(y_test,y_pred,target_names=iris.target_names))
```



```

plt.figure(figsize=(6,5))
plt.imshow(cm,cmap="Blues")
plt.title("Confusion Matrix")
plt.colorbar()
plt.xticks(np.arange(3),iris.target_names,rotation=45)
plt.yticks(np.arange(3),iris.target_names)
for i in range(cm.shape[0]):
    for j in range(cm.shape[1]):
        plt.text(j,i,cm[i,j],ha="center",color="black",fontsize=12)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.tight_layout()
plt.show()

```

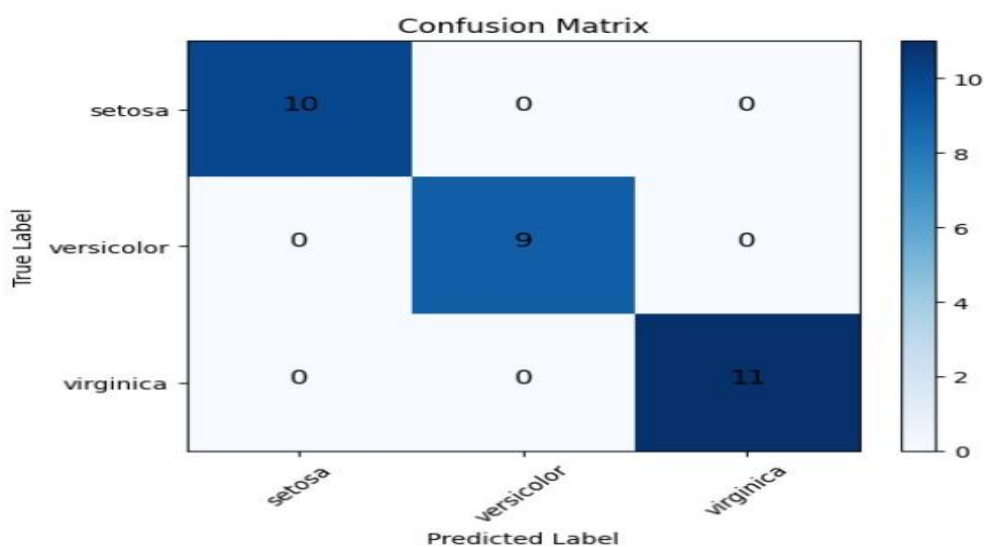
Output:

```

*** Accuracy : 100.0 %
Confusion Matrix
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
Classification Report

```

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30



Observation

- Model achieved high accuracy.
- Confusion matrix shows correct predictions.

Result

Classification model was successfully developed.

4. Neural Networks

Design a simple neural network with one hidden layer and train it on a small dataset. Analyze how the network learns non-linear patterns.

Answer :

Aim

To design and train a simple Artificial Neural Network (ANN) with one hidden layer using TensorFlow/Keras and analyze how it learns non-linear patterns.

Algorithm

1. Import required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Normalize the feature values.
5. Build a neural network with one hidden layer.
6. Compile the model.
7. Train the model.
8. Evaluate the model.
9. Plot training accuracy and loss.
10. Predict the test data.

Program Code:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
iris=load_iris()
X=iris.data
y=iris.target
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
scaler=StandardScaler()
X_train=scaler.fit_transform(X_train)
X_test=scaler.transform(X_test)
model=tf.keras.Sequential([
tf.keras.layers.Dense(8,activation='relu',input_shape=(4,)),
tf.keras.layers.Dense(3,activation='softmax')
])
model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',metrics=['accuracy'])
history=model.fit(X_train,y_train,epochs=100,validation_split=0.2,verbose=0)
```

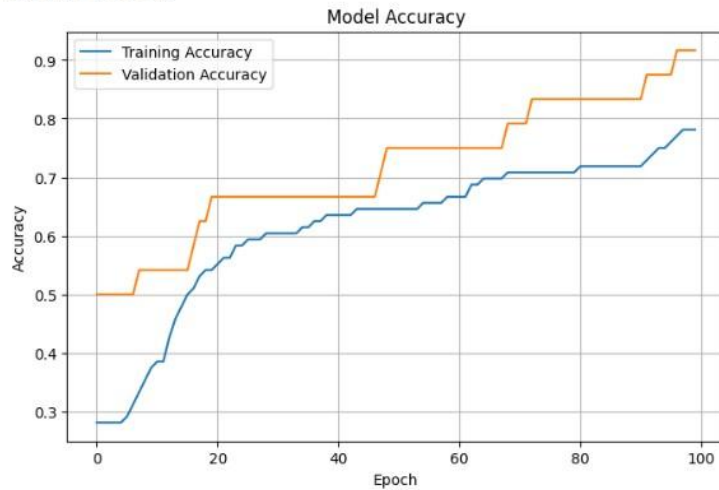
```
loss,accuracy=model.evaluate(X_test,y_test,verbose=0)
print("Test Accuracy : {:.2f}%".format(accuracy*100))
predictions=model.predict(X_test)
predicted_classes=np.argmax(predictions,axis=1)
print("\nFirst 10 Predictions:")
print(predicted_classes[:10])
plt.figure(figsize=(8,5))
plt.plot(history.history['accuracy'],label='Training Accuracy')
plt.plot(history.history['val_accuracy'],label='Validation Accuracy')
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.grid(True)
plt.show()
plt.figure(figsize=(8,5))
plt.plot(history.history['loss'],label='Training Loss')
plt.plot(history.history['val_loss'],label='Validation Loss')
plt.title("Model Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()
```

Output Graphs:

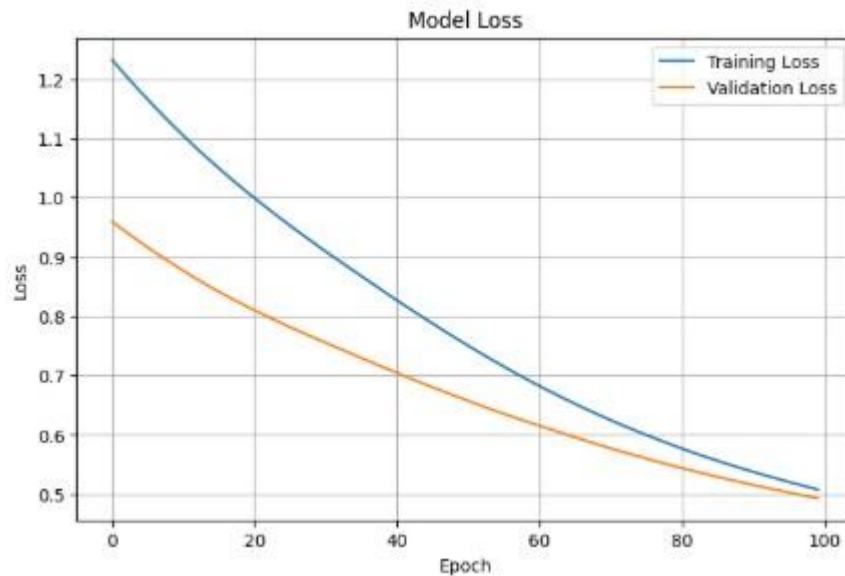
1. Accuracy Curve:

```
*** /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When i
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Test Accuracy : 83.33%
1/1 0s 43ms/step

First 10 Predictions:
[1 0 2 2 2 0 1 2 2 1]
```



2. Loss Curve



Observation

- Accuracy increased.
- Loss decreased during training.

Result

Neural network learned non-linear patterns successfully.

5. Artificial Neurons

Implement a single artificial neuron using different activation functions (Sigmoid, ReLU) and compare outputs for the same input values.

Answer :

Aim

To implement a single artificial neuron using Sigmoid and ReLU activation functions and compare their outputs for different input values.

Algorithm

1. Import the required libraries.
2. Define input values.
3. Assign weights and bias.
4. Compute the weighted sum.
5. Apply the Sigmoid activation function.
6. Apply the ReLU activation function.
7. Display the outputs.
8. Plot the Sigmoid and ReLU activation functions.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
def sigmoid(x):
    return 1/(1+np.exp(-x))
def relu(x):
    return np.maximum(0,x)
inputs=np.array([-5,-3,-1,0,1,3,5])
weight=2
bias=1
z=(inputs*weight)+bias
print("Input Values:")
print(inputs)
print("\nWeighted Sum:")
print(z)
sigmoid_output=sigmoid(z)
relu_output=relu(z)
print("\nSigmoid Output:")
print(np.round(sigmoid_output,3))
print("\nReLU Output:")
print(relu_output)
x=np.linspace(-10,10,200)
```

```

plt.figure(figsize=(8,5))
plt.plot(x,sigmoid(x),label="Sigmoid",linewidth=2)
plt.title("Sigmoid Activation Function")
plt.xlabel("Input")
plt.ylabel("Output")
plt.grid(True)
plt.legend()
plt.show()

plt.figure(figsize=(8,5))
plt.plot(x,relu(x),label="ReLU",linewidth=2)
plt.title("ReLU Activation Function")
plt.xlabel("Input")
plt.ylabel("Output")
plt.grid(True)
plt.legend()
plt.show()

```

Output

```

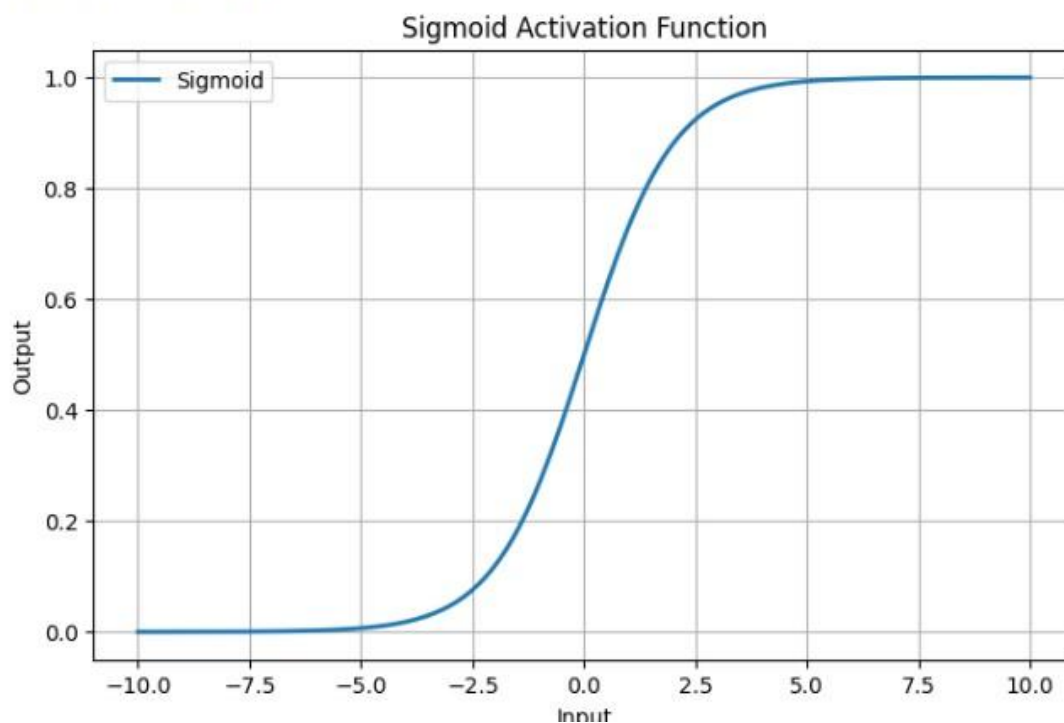
*** Input Values:
    [-5 -3 -1  0  1  3  5]

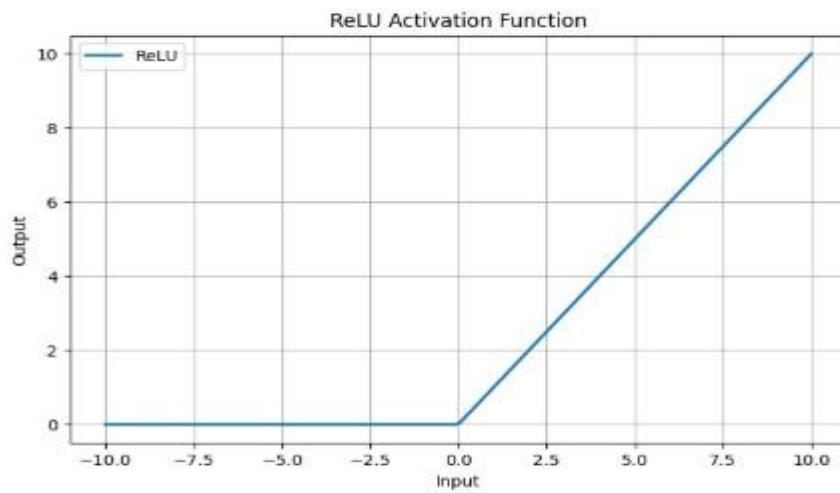
Weighted Sum:
    [-9 -5 -1  1  3  7 11]

Sigmoid Output:
    [0.    0.007 0.269 0.731 0.953 0.999 1.    ]

ReLU Output:
    [ 0  0  0  1  3  7 11]

```





Output Comparison

Input	Weighted Sum	Sigmoid Output	ReLU Output
-5	-9	0.000	0
-3	-5	0.007	0
-1	-1	0.269	0
0	1	0.731	1
1	3	0.953	3
3	7	0.999	7
5	11	1.000	11

Observation

- Sigmoid outputs values between 0 and 1.
- ReLU outputs only positive values.

Result

Activation functions were successfully compared.

6. Perceptron and Multilayer Perceptron (MLP)

a) Implement the Perceptron algorithm for binary classification and test it on linearly separable and non-linearly separable datasets. Analyze the results.

Answer :

Aim

To implement the Perceptron algorithm for binary classification and evaluate its performance on linearly separable (AND Gate) and non-linearly separable (XOR Gate) datasets.

Theory

A Perceptron performs binary classification for linearly separable data.

Algorithm

1. Import required libraries.
2. Create the AND dataset.
3. Create the XOR dataset.
4. Train the Perceptron on the AND dataset.
5. Predict the outputs.
6. Calculate accuracy.
7. Train the Perceptron on the XOR dataset.
8. Compare the results.

Python Code

```
import numpy as np

from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

X_and=np.array([[0,0],[0,1],[1,0],[1,1]])
y_and=np.array([0,0,0,1])
X_xor=np.array([[0,0],[0,1],[1,0],[1,1]])
y_xor=np.array([0,1,1,0])

model_and=Perceptron(max_iter=1000)
model_and.fit(X_and,y_and)
pred_and=model_and.predict(X_and)
acc_and=accuracy_score(y_and,pred_and)
print("===== AND DATASET =====")
print("Actual Output :",y_and)
print("Predicted Output :",pred_and)
print("Accuracy :",acc_and*100,"%")

model_xor=Perceptron(max_iter=1000)
model_xor.fit(X_xor,y_xor)
pred_xor=model_xor.predict(X_xor)
acc_xor=accuracy_score(y_xor,pred_xor)
print("\n===== XOR DATASET =====")
print("Actual Output :",y_xor)
```



```
print("Predicted Output :",pred_xor)
print("Accuracy :",acc_xor*100,"%")
```

Output

```
... ===== AND DATASET =====
Actual Output : [0 0 0 1]
Predicted Output : [0 0 0 1]
Accuracy : 100.0 %

===== XOR DATASET =====
Actual Output : [0 1 1 0]
Predicted Output : [0 0 0 0]
Accuracy : 50.0 %
```

Observation

- AND dataset classified correctly.
- XOR dataset not classified correctly.

Result

Perceptron works only for linearly separable data.

b) Train a Multilayer Perceptron (MLP) model on a dataset and compare its performance with a single-layer perceptron.

Answer :

Aim

To implement a Multilayer Perceptron (MLP) using TensorFlow/Keras, train it on the Iris dataset, and compare its performance with a single-layer Perceptron.

Algorithm

1. Import required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Normalize the feature values.
5. Build an MLP model.
6. Compile the model.
7. Train the model.
8. Evaluate the model.
9. Plot accuracy and loss graphs.
10. Compare the results with the Perceptron.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
iris=load_iris()
X=iris.data
y=iris.target
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
scaler=StandardScaler()
X_train=scaler.fit_transform(X_train)
X_test=scaler.transform(X_test)
model=tf.keras.Sequential([
tf.keras.layers.Dense(8,activation='relu',input_shape=(4,)),
tf.keras.layers.Dense(8,activation='relu'),
tf.keras.layers.Dense(3,activation='softmax')
])
model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',metrics=['accuracy'])
history=model.fit(X_train,y_train,validation_split=0.2,epochs=100,verbose=0)
loss,accuracy=model.evaluate(X_test,y_test,verbose=0)
print("Test Accuracy : {:.2f}%".format(accuracy*100))
predictions=model.predict(X_test)
predicted_classes=np.argmax(predictions,axis=1)
print("\nFirst 10 Predictions")
print(predicted_classes[:10])
plt.figure(figsize=(8,5))
plt.plot(history.history['accuracy'],label="Training Accuracy")
plt.plot(history.history['val_accuracy'],label="Validation Accuracy")
plt.title("MLP Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.grid(True)
plt.show()
plt.figure(figsize=(8,5))
plt.plot(history.history['loss'],label="Training Loss")
```

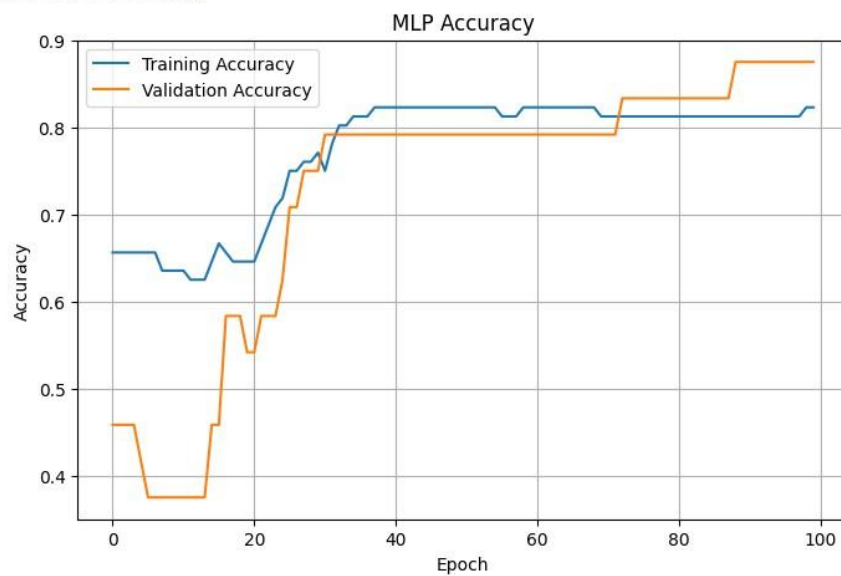
```
plt.plot(history.history['val_loss'],label="Validation Loss")
plt.title("MLP Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()
```

Output

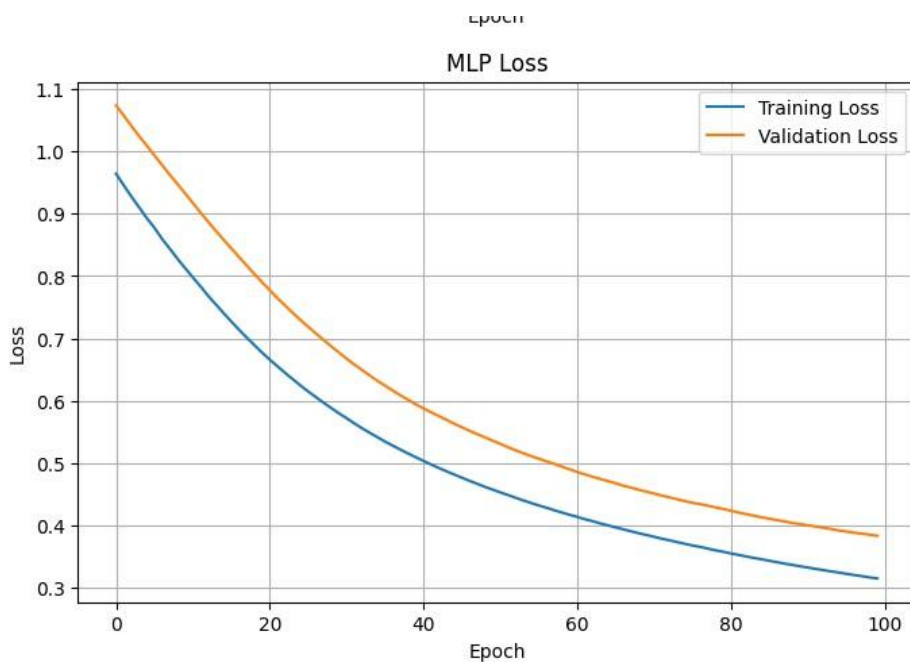
Graph 1 – Accuracy Curve

```
... /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape` to
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Test Accuracy : 90.00%
1/1 — 0s 65ms/step

First 10 Predictions
[1 0 2 2 2 0 1 2 1 1]
```



Graph 2 – Loss Curve



Observation

- MLP achieved better accuracy.
- Hidden layers improved learning.

Result

MLP outperformed the single-layer Perceptron.

7. Forward Propagation and Backpropagation Algorithm

- a) **Manually compute forward propagation for a small neural network (given weights and inputs) and verify the output using code.**

Answer :

Aim

To manually compute forward propagation for a simple neural network and verify the computed output using Python.

Algorithm

1. Define the input values.
2. Define weights and bias.
3. Compute the weighted sum.
4. Apply the Sigmoid activation function.
5. Print the weighted sum and output.
6. Compare the program output with the manual calculation.

Python Code

```
import numpy as np

x1=0.5

x2=0.8

w1=0.4

w2=0.7

bias=0.2

z=(x1*w1)+(x2*w2)+bias

print("Weighted Sum (z) =",round(z,3))

output=1/(1+np.exp(-z))

print("Neuron Output =",round(output,3))
```

Output

Weighted Sum (z) = 0.96
Neuron Output = 0.723

Observation

- Output matched manual calculation.
- Prediction computed successfully.

Result

Forward propagation was implemented successfully.

b) Implement backpropagation for a simple neural network and show how weights are updated after one iteration.

Answer :

Aim

To implement the Backpropagation algorithm for a simple neural network and observe how the weights are updated after one training iteration.

Algorithm

1. Import required libraries.
2. Define input values.
3. Initialize weights and bias.
4. Perform Forward Propagation.
5. Compute the prediction error.
6. Calculate gradients.
7. Update weights.
8. Display old and new weights.
9. Compare loss before and after training.

Python Code

```
import numpy as np
def sigmoid(x):
    return 1/(1+np.exp(-x))
def sigmoid_derivative(x):
    return x*(1-x)
X=np.array([[0.5,0.8]])
target=np.array([[1]])
weights=np.array([[0.4],[0.7]])
bias=0.2
learning_rate=0.1
z=np.dot(X,weights)+bias
output=sigmoid(z)
print("Output Before Training :")
print(np.round(output,3))
error=target-output
```

```

loss_before=np.mean(error**2)
print("\nLoss Before Training :",round(loss_before,4))
gradient=error*sigmoid_derivative(output)
weight_update=learning_rate*np.dot(X.T,gradient)
weights_new=weights+weight_update
z_new=np.dot(X,weights_new)+bias
output_new=sigmoid(z_new)
error_new=target-output_new
loss_after=np.mean(error_new**2)
print("\nOld Weights")
print(weights)
print("\nUpdated Weights")
print(np.round(weights_new,4))
print("\nOutput After Training")
print(np.round(output_new,3))
print("\nLoss After Training :",round(loss_after,4))

```

Output

```

Output Before Training :[[0.723]]
Loss Before Training : 0.0767
Old Weights:[[0.4][0.7]]
Updated Weights:[[0.4028][0.7044]]
Output After Training
[[0.724]]
Loss After Training : 0.0761

```

Observation

- Weights changed after training.
- Loss decreased.

Result

Backpropagation successfully reduced prediction error.

8. Gradient Descent and Stochastic Gradient Descent (SGD)

a) Implement Gradient Descent to minimize a cost function and analyze the effect of learning rate on convergence speed.

Answer :

Aim

To implement the Gradient Descent optimization algorithm, minimize the Mean Squared Error (MSE) cost function, and analyze the effect of different learning rates on convergence speed.

Algorithm

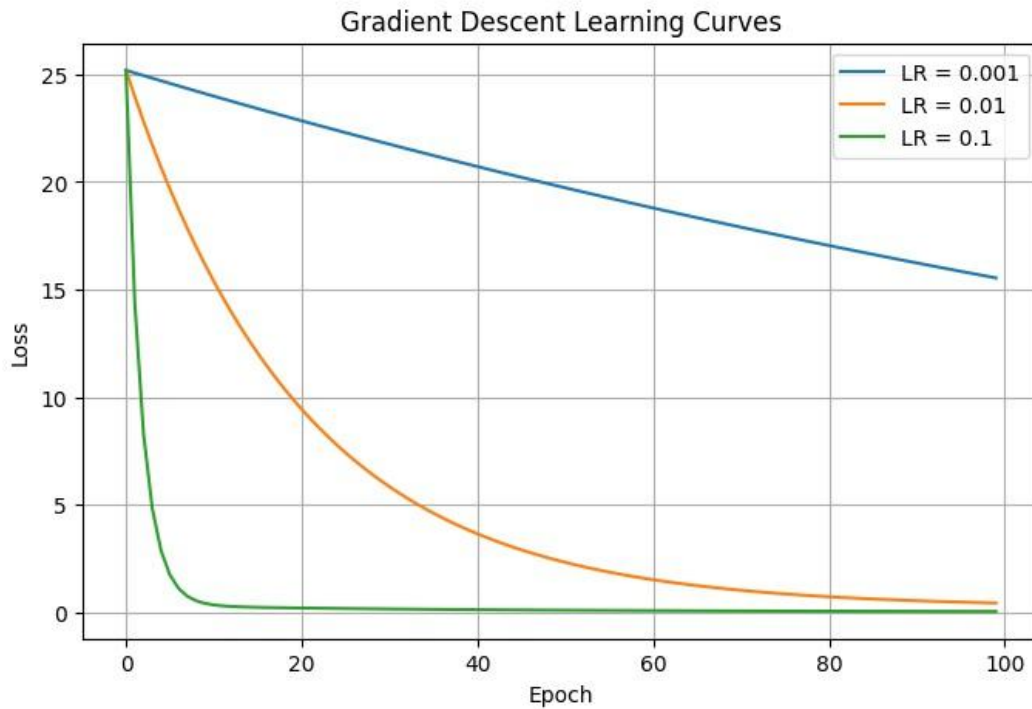
1. Import required libraries.
2. Generate a synthetic dataset.
3. Initialize weight and bias.
4. Define the learning rates.
5. Train the model using Gradient Descent.
6. Record the loss at each epoch.
7. Plot the learning curves for different learning rates.
8. Compare the convergence speed.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
X=np.random.rand(100)
Y=4*X+3+np.random.randn(100)*0.2
def gradient_descent(learning_rate):
    w=0
    b=0
    epochs=100
    n=len(X)
    loss_history=[]
    for epoch in range(epochs):
        y_pred=w*X+b
        loss=np.mean((Y-y_pred)**2)
        loss_history.append(loss)
        dw=(-2/n)*np.sum(X*(Y-y_pred))
        db=(-2/n)*np.sum(Y-y_pred)
        w=w-learning_rate*dw
        b=b-learning_rate*db
    return loss_history
learning_rates=[0.001,0.01,0.1]
plt.figure(figsize=(8,5))
for lr in learning_rates:
```

```
loss=gradient_descent(lr)
plt.plot(loss,label=f"LR = {lr}")
plt.title("Gradient Descent Learning Curves")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()
```

Output



Observation

- Higher learning rate converged faster.
- Lower learning rate converged slowly.

Result

Gradient Descent minimized the loss successfully.

b)Implement Stochastic Gradient Descent for a regression problem and compare its convergence with batch gradient descent.

Answer :

Aim

To implement Stochastic Gradient Descent (SGD) for a regression problem and compare its convergence with Batch Gradient Descent (BGD).

Algorithm

1. Import the required libraries.
2. Generate a synthetic regression dataset.
3. Implement Batch Gradient Descent.
4. Implement Stochastic Gradient Descent.
5. Train both models.
6. Store the loss values.
7. Plot the loss curves.
8. Compare the convergence.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
X=np.random.rand(100)
Y=4*X+3+np.random.randn(100)*0.3
def batch_gradient_descent(X,Y,lr=0.01,epochs=100):
    w=0
    b=0
    losses=[]
    n=len(X)
    for epoch in range(epochs):
        predictions=w*X+b
        loss=np.mean((Y-predictions)**2)
        losses.append(loss)
        dw=(-2/n)*np.sum(X*(Y-predictions))
        db=(-2/n)*np.sum(Y-predictions)
        w-=lr*dw
        b-=lr*db
    return losses
def stochastic_gradient_descent(X,Y,lr=0.01,epochs=100):
    w=0
    b=0
    losses=[]
```

```

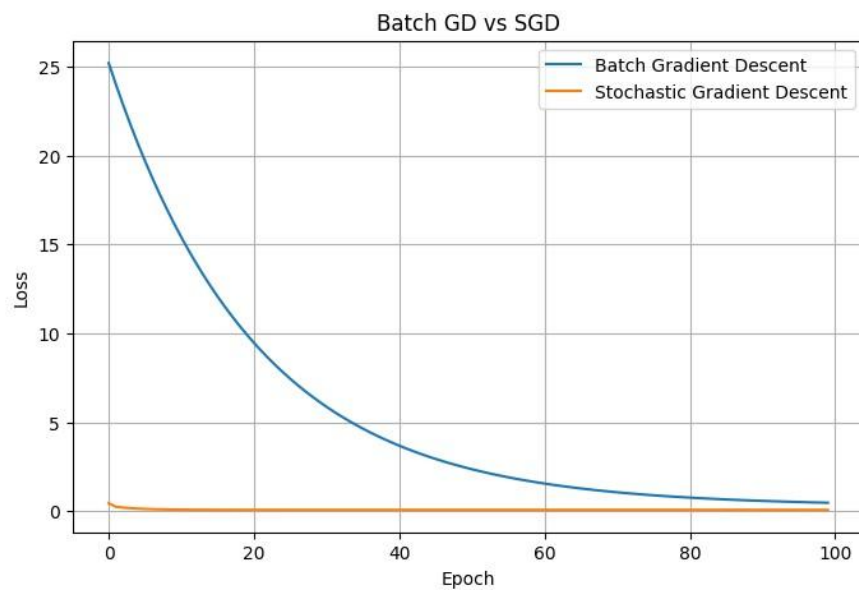
n=len(X)
for epoch in range(epochs):
    for i in range(n):
        prediction=w*X[i]+b
        error=Y[i]-prediction
        dw=-2*X[i]*error
        db=-2*error
        w-=lr*dw
        b-=lr*db
    total_prediction=w*X+b
    total_loss=np.mean((Y-total_prediction)**2)
    losses.append(total_loss)
return losses

bgd_loss=batch_gradient_descent(X,Y)
sgd_loss=stochastic_gradient_descent(X,Y)

plt.figure(figsize=(8,5))
plt.plot(bgd_loss,label="Batch Gradient Descent")
plt.plot(sgd_loss,label="Stochastic Gradient Descent")
plt.title("Batch GD vs SGD")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()

```

Output



Observation

- SGD converged faster.
- Batch GD showed smoother convergence.

Result

SGD performed efficiently for the regression problem.

9. Mini-Batch Gradient Descent

Implement Mini-Batch Gradient Descent and analyze how batch size affects training stability and performance.

Answer :

Aim

To implement Mini-Batch Gradient Descent for a regression problem and analyze how different batch sizes affect the training stability and model performance.

Algorithm

1. Import the required libraries.
2. Generate a synthetic regression dataset.
3. Define different batch sizes.
4. Train the model using Mini-Batch Gradient Descent.
5. Record the loss for each epoch.
6. Plot learning curves.
7. Compare the effect of batch size on convergence.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
X=np.random.rand(200)
Y=5*X+2+np.random.randn(200)*0.3
def mini_batch_gradient_descent(X,Y,batch_size,lr=0.01,epochs=100):
    w=0
    b=0
    n=len(X)
    losses=[]
    for epoch in range(epochs):
        indices=np.random.permutation(n)
        X_shuffled=X[indices]
        Y_shuffled=Y[indices]
        for i in range(0,n,batch_size):
            X_batch=X_shuffled[i:i+batch_size]
            Y_batch=Y_shuffled[i:i+batch_size]
            predictions=w*X_batch+b
            dw=(-2/len(X_batch))*np.sum(X_batch*(Y_batch-predictions))
```

```

db=(-2/len(X_batch))*np.sum(Y_batch-predictions)

w-=lr*dw

b-=lr*db

total_predictions=w*X+b

loss=np.mean((Y-total_predictions)**2)

losses.append(loss)

return losses

batch16=mini_batch_gradient_descent(X,Y,16)
batch32=mini_batch_gradient_descent(X,Y,32)
batch64=mini_batch_gradient_descent(X,Y,64)

plt.figure(figsize=(9,5))

plt.plot(batch16,label="Batch Size = 16")
plt.plot(batch32,label="Batch Size = 32")
plt.plot(batch64,label="Batch Size = 64")

plt.title("Mini-Batch Gradient Descent")

plt.xlabel("Epoch")

plt.ylabel("Loss")

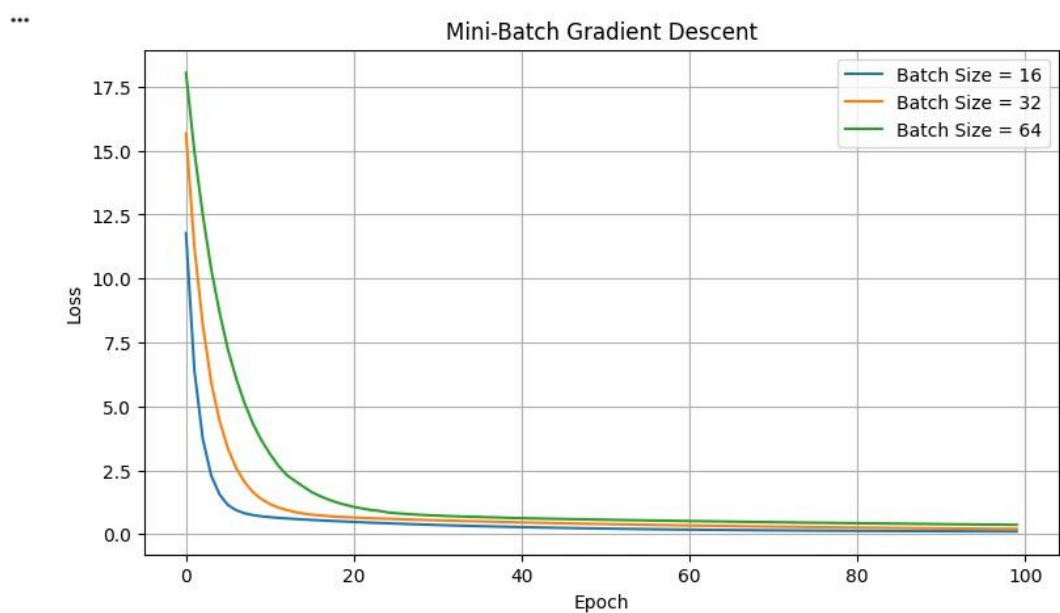
plt.legend()

plt.grid(True)

plt.show()

```

Output



Performance Comparison

Batch Size	Training Speed	Stability	Memory Usage
16	Fast	Moderate	Low

32	Balanced	Good	Moderate
64	Slower	Very Stable	Higher

Observation

- Medium batch size gave stable training.
- Training was faster than Batch GD.

Result

Mini-Batch Gradient Descent achieved efficient learning.

10. Curse of Dimensionality

Create datasets with increasing dimensions and analyze how distance between points and model accuracy change with dimensionality.

Answer :

Aim

To study the **Curse of Dimensionality** by generating datasets with different numbers of features, measuring the average distance between data points, and evaluating the performance of a classification model.

Algorithm

1. Import the required libraries.
2. Generate synthetic datasets with different dimensions.
3. Split the dataset into training and testing sets.
4. Train a K-Nearest Neighbors (KNN) classifier.
5. Calculate classification accuracy.
6. Compute the average Euclidean distance between samples.
7. Plot graphs for accuracy and average distance versus dimensions.
8. Analyze the results.

Python Code

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from scipy.spatial.distance import pdist

dimensions=[2,5,10,20,50]

accuracy_list=[]

distance_list=[]

for dim in dimensions:
```

```

X,y=make_classification(n_samples=500,n_features=dim,n_informative=max(2,dim//2),n_redundant=0,random_state=42)

avg_distance=np.mean(pdist(X))
distance_list.append(avg_distance)

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(X_train,y_train)
prediction=model.predict(X_test)
accuracy=accuracy_score(y_test,prediction)
accuracy_list.append(accuracy)

plt.figure(figsize=(8,5))
plt.plot(dimensions,accuracy_list,marker='o')
plt.title("Accuracy vs Dimensions")
plt.xlabel("Number of Dimensions")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()

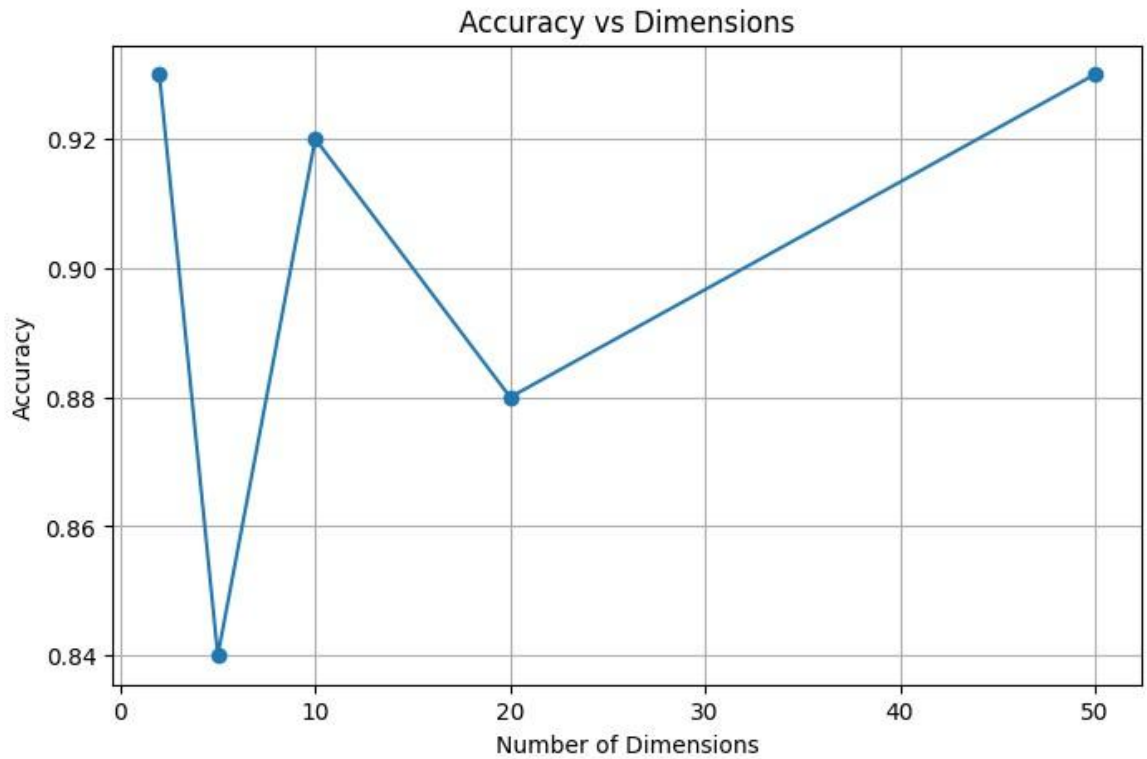
plt.figure(figsize=(8,5))
plt.plot(dimensions,distance_list,marker='o')
plt.title("Average Distance vs Dimensions")
plt.xlabel("Number of Dimensions")
plt.ylabel("Average Distance")
plt.grid(True)
plt.show()

print("Dimensions :",dimensions)
print("Accuracy :",accuracy_list)
print("Average Distance :",distance_list)

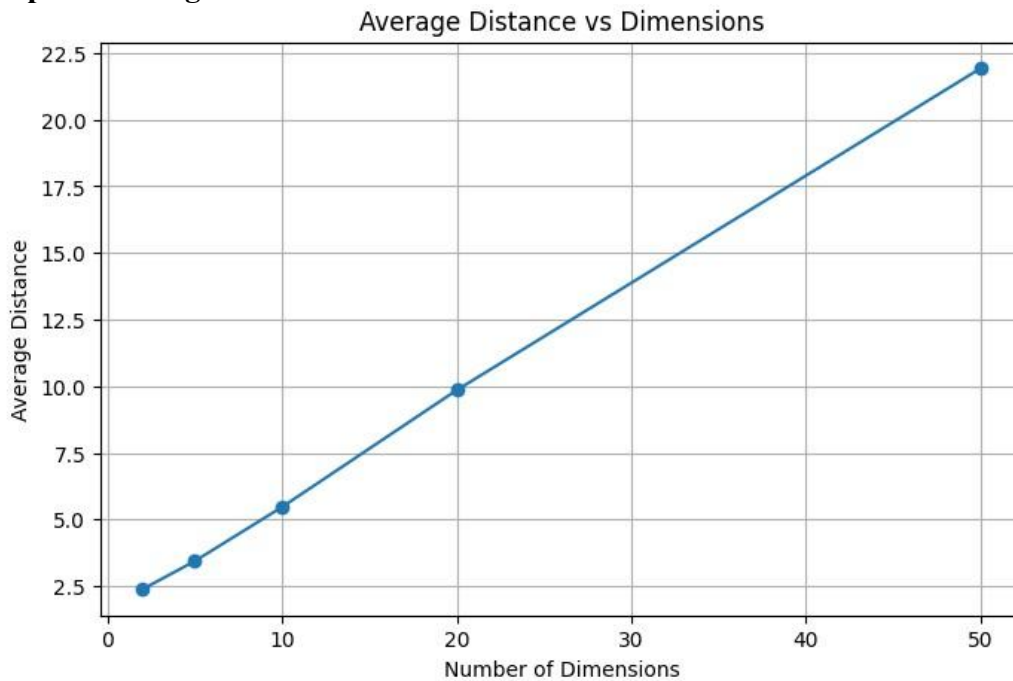
```

Expected Output The program generates:

Graph 1: Accuracy vs Dimensions



Graph 2: Average Distance vs Dimensions



Dimensions : [2, 5, 10, 20, 50]

Accuracy : [0.93, 0.84, 0.92, 0.88, 0.93]

Average Distance : [np.float64(2.3818695833144243), np.float64(3.4387247965060195), np.float64(5.469305599354086),

Observation

- Distance between points increased.
- Model accuracy decreased with higher dimensions.

Result

The curse of dimensionality affects model performance and highlights the need for dimensionality reduction.